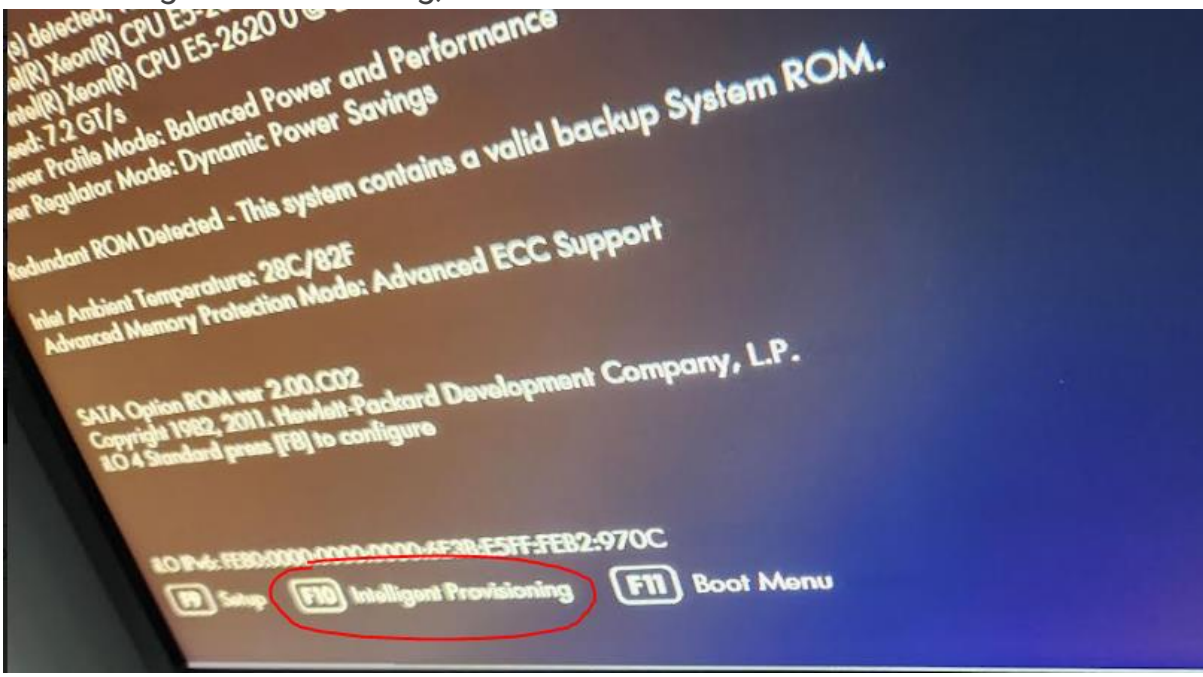
1. Download [the latest version of SystemRescue](#).

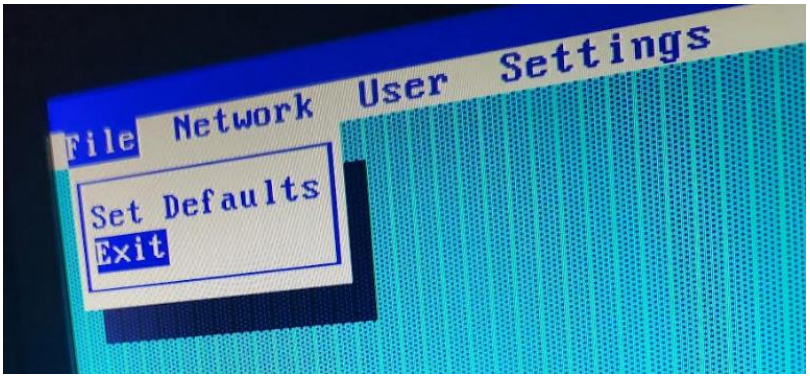
2. Create a USB bootable from the downloaded iso. On Microsoft Windows, one could use [Rufus](#). Or, the cross-platform free USB flash tool for Microsoft Windows, macOS, and Linux, [balenaEtcher](#).
3. Insert the sd card inside the DL360 server SD Card Slot. Ensure that SD Card is empty.

4. Before starting SystemRescue from the USB drive, make sure to delete all the RAID configurations on the controller by pressing F8 at boot time. This is mandatory! And, make sure to make a backup of everything for obvious reasons, otherwise, your data will be wiped!
    a. Power on the DL360,
    b. F10 intelligent Provisioning,



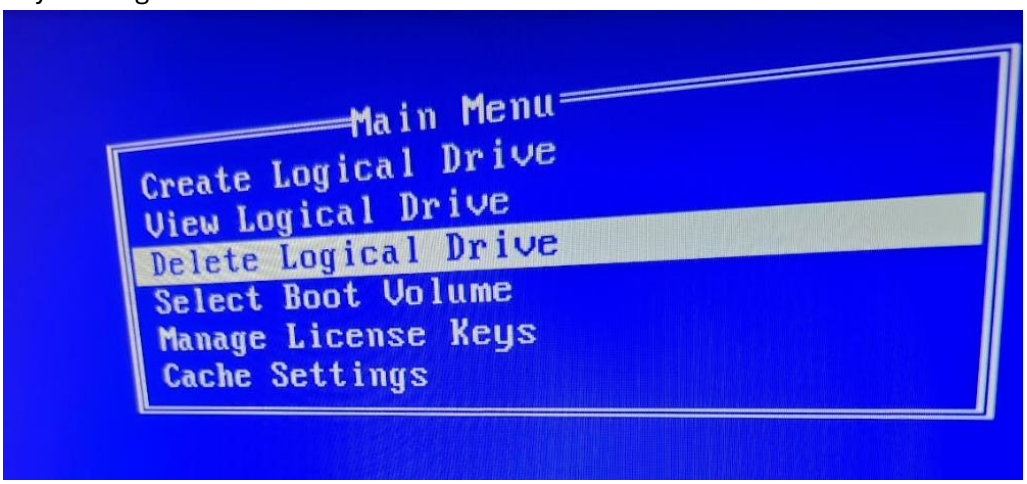
    c. then f8,

d. Exit this screen if it comes up

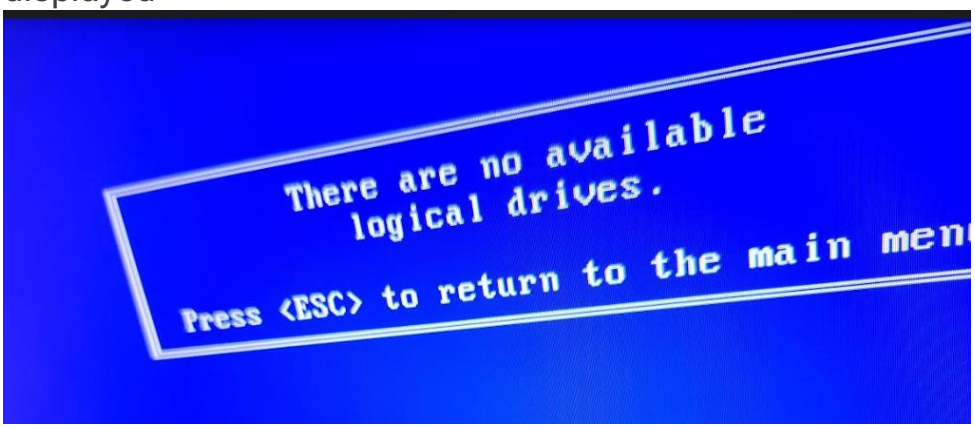


e. Then F8 again

f. Select the Delete Logical Drive , and follow the on screen instruction to delete any existing drive



g. Once everything has been deleted, this page should now displayed



5. Insert the USB that has the systemrescue

a. Reboot the server and then do F11 for Boot menu and then 3 One time boot to USB DriveKey
b. In the system rescue menu, select SystemRescue using default options.
c. after booting SystemRescue you could type startx in order to start a neat and clean Xfce session.

6. Format the sd card

   open the terminal, and run command:

   a. fsdisk -l
   b. wipefs -a /dev/sdj replace the j with the letter the system assigned to the sd card

7. open the terminal, and run command: sudo pacman -Syu to update packman
8. command: Pacman -Sy
9. command: sudo pacman -S --needed base-devel git wget yajl
10. command: nano /etc/passwd

    a. change

       nobody:x:65534:65534:KernelOverflowser:/:/usr/bin/nologin to **nobody:x:65534:65534:Kernel Overflow User:/:/bin/bash** then save

11. commands below:

    a. cd /var/tmp
    b. su nobody -c 'git clone https://aur.archlinux.org/ssacli.git'
    c. cd ssacli
    d. su nobody -c 'makepkg -si'
    e. cp -vr pkg/ssacli/opt /
    f. cp -vr pkg/ssacli/usr/share /usr/
    g. cp -vr pkg/ssacli/usr/bin/* /usr/bin/

h. hash -r

12. Finally, it's time to enable the HBA mode (pass-through)

a. ssacli controller slot=0 modify hbamode=on

you can re-enable hardware RAID and disable HBA mode at any time by starting over from step `1` and only set `hbamode=off` instead of `hbamode=on`.

13. Confirm the HBA mode has been enabled:

a. ssacli controller slot=0 show | grep -i hba
b. ssacli controller slot=0 show

14. We're done! Poweroff the server:

15. Remove systemrescue USB

16. Insert the proxmox bootable USB drive in the server USB Ports

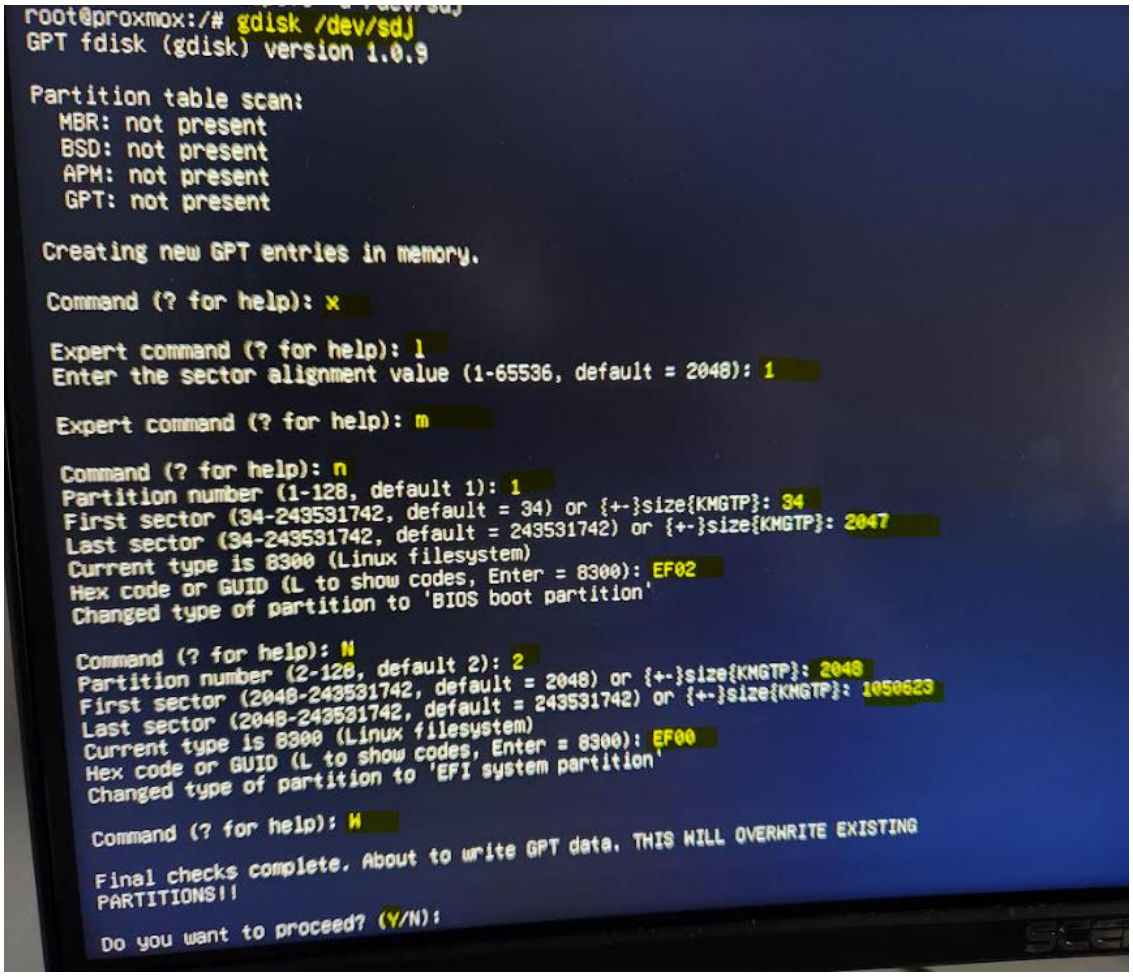
17. Poweron the server

18. install proxmox in my case I installed it as ZFS Raid1 but **uncheck** the flag "automatically reboot successful after installation".

19. When the installation is done, instead of clicking reboot, hold CTRL+ALT+F3

20. fsdisk -a my sd card was sdj

21. gdisk /dev/sdj



22. lsblk command show a new partition name sdj1 and sdj2
23. Now mount rpool to /mnt with `zpool import rpool -R /mnt`
    a. mount -t proc proc /mnt/proc
    b. mount -t sysfs sys /mnt/sys
    c. mount -o bind /dev /mnt/dev
    d. mount -o bind /run /mnt/run
24. `chroot /mnt`
25. if encounter error about "contains a file systems vfat" when running boot tool command ,try command: fdisk /dev/sdj then take the GPT option, then type **quit** to exit. Then below again
    a. proxmox-boot-tool format /dev/sdj2 --force and proxmox-boot-tool init /dev/ sdj2
26. Now check if the UUID of /dev/sdj2 from blkid /dev/sdj2 is inside `/etc/kernel/proxmox-boot-uuids`, if not try to run proxmox-boot-tool refresh
27. `zpool export rpool`, not needed
28. ctrl+alt+f4
29. now click reboot
30. remove proxmox usb installer

credits to :
https://www.babaei.net/blog/disable-hp-proliant-ml350p-gen8-p420i-raid-controller-enable-hba-mode-pass-through-perform-freebsd-root-zfs-installation/
https://forum.proxmox.com/threads/move-boot-partition-zfs-usb.137870/

https://www.reddit.com/r/homelab/comments/ap9usf/proxmoxzfs_installed_on_hp_dl360p/

https://bbs.archlinux.org/viewtopic.php?id=198066

https://unix.stackexchange.com/questions/650001/move-boot-partition-of-proxmox-or-debian-to-usb-flash-installed-on-two-nvme-d/662861#662861

https://www.linuxquestions.org/questions/linux-newbie-8/mount-point-does-not-exist-~-trying-to-mount-a-nas-4175685417/

https://forum.proxmox.com/threads/booting-proxmox-from-internal-sd-card-only-boot-partition.94957/